# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

If $f_r(x), g_r(x), h_r(x), r = 1, 2, 3$ are polynomials in $x$ such that $f_r(a) = g_r(a) = h_r(a), r = 1, 2, 3$ and $F(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$ then $F'(a)$ is

**(A)** - 1

**(B)** $a$

**(C)** 0

**(D)** none of these

**Q.2**

Let $A\left(x_1, y_1\right), B\left(x_2, y_2\right)$ and $C\left(x_3, y_3\right)$ be vertices of an equilateral triangle whose side is 4 units. Let $\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ then $\Delta^2$ is equal to

**(A)** 64

**(B)** 128

**(C)** 192

**(D)** 256

**Q.3**

If $\Delta = \begin{vmatrix} a+bx & c+dx & p+qx \\ ax+b & cx+d & px+q \\ u & v & w \end{vmatrix} = \lambda \begin{vmatrix} a & c & p \\ b & d & q \\ u & v & w \end{vmatrix}$ then $\lambda$ equals

**(A)** 0

**(B)** 1

**(C)** $x$

**(D)** $1 - x^2$

**Q.4**

If $\alpha, \beta, \gamma$ are real numbers, then the determinant $\Delta = \begin{vmatrix} \sin^2\alpha & \cos 2\alpha & \cos^2\alpha \\ \sin^2\beta & \cos 2\beta & \cos^2\beta \\ \sin^2\gamma & \cos 2\gamma & \cos^2\gamma \end{vmatrix}$ equals

**(A)** 0

**(B)** $-1$

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**(C)** $\sin^2\alpha+\sin^2\beta+\sin^2\gamma$

**(D)** none of these

**Q.5**

If $\mathrm{D}$ is determinant of order 3 and $\mathrm{D}'$ is the determinant obtained by replacing the elements of $D$ by their cofactors, then which one of the following is correct?

**(A)** $D'=D^2$

**(B)** $D'=D^3$

**(C)** $D'=2D^2$

**(D)** $\mathrm{D}'=3\mathrm{D}^3$

**Q.6**

If $(\mathrm{x}),\mathrm{g}(\mathrm{x})$ and $\mathrm{h}(\mathrm{x})$ are three polynomials of degree 2and $\Delta(x)=\begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}$, then $\Delta(x)$ is polynomial of degree

**(A)** 2

**(B)** 3

**(C)** at most 2

**(D)** at most 3

**Q.7**

Let $\Delta=\begin{vmatrix} \sin x & \sin(x+h) & \sin(x+2h) \\ \sin(x+2h) & \sin x & \sin(x+h) \\ \sin(x+h) & \sin(x+2h) & \sin x \end{vmatrix}$

Then, $\lim_{\mathrm{h}\to 0}\left(\frac{\Delta}{\mathrm{h}^2}\right)$ is

**(A)** $9\sin^2 x\cos x$

**(B)** $3\cos^2 x$

**(C)** $\sin x\cos^2 x$

**(D)** None of these

**Q.8**

If $y=\begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix}$, then $\frac{dy}{dx}$ is

**(A)** 1

**(B)** 2

**(C)** 3

**(D)** 0

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**Q.9**

If $\Delta_r = \begin{vmatrix} r-1 & n & 6 \\ (r-1)^2 & 2n^2 & 4n-2 \\ (r-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$ then $\sum_{r=1}^{n-1} \Delta_r$ is.

**(A)** 0

**(B)** 1

**(C)** 3

**(D)** $-1$

**Q.10**

If $\Delta_r = \begin{vmatrix} 2r-1 & {}^mC_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix}$ then value of $\sum_{r=0}^{m} \Delta_r$ is

**(A)** 0

**(B)** 1

**(C)** 2

**(D)** 5

**Q.11**

For all values of $A, B, C$ and $P, Q, R$ the value of the determinant $(x+a)^3 \begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix}$ is

**(A)** 1

**(B)** 0

**(C)** 2

**(D)** None of these

**Q.12**

The value of $\Delta = \begin{vmatrix} x^2+y^2 & ax+by & px+qy \\ ax+by & a^2+b^2 & ap+bq \\ px+q^y & ap+bq & p^2+q^2 \end{vmatrix}$ is equal to

**(A)** $p+q$

**(B)** $a+b+c$

**(C)** $x+y+z$

**(D)** 0

**Q.13**

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Let $S_k = \alpha^k + \beta^k + \gamma^k$, then

$\Delta = \begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix}$ is equal to

**(A)** $S_6$

**(B)** $S_5 - S_3$

**(C)** $S_6 - S_4$

**(D)** None of these

**Q.14**

Let $A, B$ and $C$ be $n \times n$ matrices. Which one of the following is a correct statement?

**(A)** If $AB = AC$, then $B = C$

**(B)** If $A^3 + 2A^2 + 3A + 5I = 0$; then $A$ is invertible.

**(C)** If $A^2 = 0$, then $A = 0$

**(D)** None of these

**Q.15**

Let $F(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$ where $\alpha \in \mathbf{R}$. Then $[F(\alpha)]^{-1}$ is equal to

**(A)** $F(-\alpha)$

**(B)** $F\left(\alpha^{-1}\right)$

**(C)** $F(2\alpha)$

**(D)** None of these

## Answers & Solutions

**Q.1 Answer:**

0

**Solution:**

**Differentiating w.r.t. $x$ we get $F'(x)$**

$$\begin{vmatrix} f_1'(x) & f_2'(x) & f_3'(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1'(x) & g_2'(x) & g_3'(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1'(x) & h_2'(x) & h_3'(x) \end{vmatrix}$$

**Since $f_r(a) = g_r(a) = h_r(a)$ for $r = 1, 2, 3$, we get each of the three determinants on the right side becomes zero when $x$ is replaced by $a$. (In each case two rows become identical.) Thus, $F'(a) = 0$.**

**Q.2 Answer:**

192

**Solution:**

$\frac{1}{2}|\Delta| =$ **area of triangle** $= \frac{\sqrt{3}}{4}(4)^2 = 4\sqrt{3}$

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$\Rightarrow \quad |\Delta| = 8\sqrt{3} \Rightarrow \Delta^2 = 192$

**Q.3 Answer:**

$1 - x^2$

**Solution:**

**Applying $R_1 \to R_1 - xR_2$ we get**

$$\Delta = \begin{vmatrix} a\left(1-x^2\right) & c\left(1-x^2\right) & p\left(1-x^2\right) \\ ax+b & cx+d & px+q \\ u & v & w \end{vmatrix}$$

**Take $1 - x^2$ common from $R_1$ and apply $R_2 \to R_2 - xR_1$ to obtain**

$$\Delta = \left(1-x^2\right)\begin{vmatrix} a & c & p \\ b & d & q \\ u & v & w \end{vmatrix}$$

**Q.4 Answer:**

none of these

**Solution:**

**We know**

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$\text{Let } \Delta_1 = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

**Using $R_3 \to R_3 - R_2$ and $R_2 \to R_2 - R_1$, we get**

$$\Delta_1 = \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b^2-a^2 & b^3-a^3 \\ 0 & c^2-b^2 & c^3-b^3 \end{vmatrix}$$

$$= (b-a)(c-b)\begin{vmatrix} b+a & b^2+a^2+ba \\ c+b & c^2+b^2+cb \end{vmatrix}$$

**Applying $C_2 \to C_2 - bC_1$, we get**

$$\Delta_1 = (b-a)(c-b)\begin{vmatrix} b+a & a^2 \\ c+b & c^2 \end{vmatrix}$$

$$= (b-a)(c-b)\begin{vmatrix} b+a & a^2 \\ c-a & c^2-a^2 \end{vmatrix}$$

$$[\text{using } R_2 \to R_2 - R_1]$$

$$= (b-a)(c-b)(c-a)\begin{vmatrix} b+a & a^2 \\ 1 & c+a \end{vmatrix}$$

$$= (a-b)(b-c)(c-a)(bc+ca+ab)$$

$$\text{Thus,} \quad \lambda = 18$$

**Q.5 Answer:**

$D' = D^2$

**Solution:**

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$$
\begin{aligned}
&\mathrm{D}' = \text{ cofactor D} \\
&\Rightarrow |\mathrm{D}'| = |\text{ cofactor D }| \\
&\Rightarrow |\mathrm{D}'| = |\mathrm{D}|^{3-1}. \\
&\Rightarrow |\mathrm{D}'| = |\mathrm{D}|^2 \\
&\text{So } \mathrm{D}' = \mathrm{D}^2
\end{aligned}
$$

**Q.6 Answer:**

at most 2

**Solution:**

**Let** $\mathrm{f(x)} = \mathrm{a_0x^2 + a_1x + a_2}$

$g(x) = b_0x^2 + b_1x + b_2$

$\mathrm{h(x)} = \mathrm{c_0x^2 + c_1x + c_2}$

**Then,** $\Delta(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ 2a_0x + a_1 & 2b_0x + b_1 & 2c_0x + c_1 \\ 2a_0 & 2b_0 & 2c_0 \end{vmatrix}$

$= \mathrm{x}\begin{vmatrix} f(x) & g(x) & h(x) \\ 2a_0 & 2b_0 & 2c_0 \\ 2a_0 & 2b_0 & 2c_0 \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ a_1 & b_1 & c_1 \\ 2a_0 & 2b_0 & 2c_0 \end{vmatrix}$

$= 0 + 2\begin{vmatrix} f(x) & g(x) & h(x) \\ a_1 & b_1 & c_1 \\ a_0 & b_0 & c_0 \end{vmatrix}$

$= 2\left[(b_1c_0 - b_0c_1)\, f(x) - (a_1c_0 - a_0c_1)\, g(x)\right.$

$\left. + (a_1b_0 - a_0b_1)\, h(x)\right]$

**Hence degree of** $\Delta(\mathrm{x}) \leq 2$

**Q.7 Answer:**

$3\cos^2 x$

**Solution:**

**Operating** $\mathrm{C_1 \to C_1 + C_2 + C_3}$ **and taking**

**out the common factor from** $\mathrm{C_1}$**, we have** $\Delta = \sin(x+h)(1 + 2\cos h)$

$\begin{vmatrix} 1 & \sin(x+h) & \sin(x+2h) \\ 1 & \sin x & \sin(x+h) \\ 1 & \sin(x+2h) & \sin x \end{vmatrix}$

{ **Since,** $\sin(x+h) + \sin(x) + \sin(x+2h)$

$= \sin(x+h)(1 + 2\cos h)$

**By operating** $\mathrm{R_2 \to R_2 - R_1}$ **and** $\mathrm{R_3 \to R_3 - R_1}$**,**

**we have** $\Delta = \sin(x+h)(1 + 2\cos h)$

$\begin{vmatrix} 1 & \sin(x+h) & \sin(x+2h) \\ 0 & \sin x - \sin(x+h) & \sin(x+h) - \sin(x+2h) \\ 0 & \sin(x+2h) - \sin(x+h) & \sin x - \sin(x+2h) \end{vmatrix}$

$= \left\{ \begin{array}{l} \left(2\cos\left(x + \frac{\mathrm{h}}{2}\right)\sin + \frac{\mathrm{h}}{2}\right)\left(2\cos(\mathrm{x+h})\sin \mathrm{h}\right) \\ + \left(2\cos\left(\mathrm{x} + \frac{3\,\mathrm{h}}{2}\right)\sin\frac{\mathrm{h}}{2}\right)^2 \end{array} \right\}$

**Therefore,** $\lim_{h\to 0}\frac{\Delta}{h^2} = 3\cos^2 x$

**Q.8 Answer:**

1

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**Solution:**

$$\frac{dy}{dx}=\begin{vmatrix}\cos x & -\sin x & \cos x\\ \cos x & -\sin x & \cos x\\ x & 1 & 1\end{vmatrix}$$
$$+\begin{vmatrix}\sin x & \cos x & \sin x\\ -\sin x & -\cos x & -\sin x\\ x & 1 & 1\end{vmatrix}+\begin{vmatrix}\sin x & \cos x & \sin x\\ \cos x & -\sin x & \cos x\\ 1 & 0 & 0\end{vmatrix}$$
$$=0-\begin{vmatrix}\sin x & \cos x & \sin x\\ \sin x & \cos x & \sin x\\ x & 1 & 1\end{vmatrix}+1\begin{vmatrix}\cos x & \sin x\\ -\sin x & \cos x\end{vmatrix}$$
$$=0+\left(\cos^2 x+\sin^2 x\right)$$

**Q.9 Answer:**

0

**Solution:**

**Since $C_1$ has variable terms and $C_2$ and $C_3$ are constant, summation runs on $C_1$. Therefore,** $\sum_{r=1}^{n}\Delta_r=$
$$\begin{vmatrix}\sum_1^n(r-1) & n & 6\\ \sum_1^n(r-1)^2 & 2n^2 & 4n-2\\ \sum_1^n(r-1)^3 & 3n^2 & 3n^2-3n\end{vmatrix}$$
$$=\begin{vmatrix}\frac{1}{2}(n-1)n & n & 6\\ \frac{1}{6}(n-1)n(2n-1) & 2n^2 & 4n-2\\ \frac{1}{4}(n-1)^2n^2 & 3n^3 & 3n^2-3n\end{vmatrix}$$
**Taking $\frac{1}{12}n(n-1)$ common from $C_1$ and n common from $C_2$, we get**
$$\sum\Delta_r=\frac{1}{12}n^2(n-1)\times\begin{vmatrix}6 & 1 & 6\\ 2(2n-1) & 2n & 2(2n-1)\\ 3n(n-1) & 3n^3 & 3n(n-1)\end{vmatrix}$$
$$=0\quad[\because C_1 \text{ and } C_3 \text{ are identical}]$$

**Q.10 Answer:**

0

**Solution:**

$$\Delta_r=\begin{vmatrix}2r-1 & {}^mC_r & 1\\ m^2-1 & 2^m & m+1\\ \sin^2\left(m^2\right) & \sin^2(m) & \sin^2(m+1)\end{vmatrix}$$
$$\therefore\sum_{r=0}^{m}\Delta_r=\begin{vmatrix}\sum_{r=0}^{m}(2r-1) & \sum_{r=0}^{m}{}^mC_r & \sum_{r=0}^{m}1\\ m^2-1 & 2^m & m+1\\ \sin^2\left(m^2\right) & \sin^2(m) & \sin^2(m+1)\end{vmatrix}$$
$$=\begin{vmatrix}m^2-1 & 2^m & m+1\\ m^2-1 & 2^m & m+1\\ \sin^2\left(m^2\right) & \sin^2(m) & \sin^2(m+1)\end{vmatrix}$$
$$=0(\because \text{two rows are identical})$$
**Hence, option B is correct.**

**Q.11 Answer:**

0

**Solution:**

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**We have,** $\begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix}$

$\begin{vmatrix} \cos A & \sin A & 0 \\ \cos B & \sin B & 0 \\ \cos C & \sin C & 0 \end{vmatrix} \times \begin{vmatrix} \cos P & \sin P & 0 \\ \cos Q & \sin Q & 0 \\ \cos R & \sin R & 0 \end{vmatrix} = 0 \cdot 0 = 0$

**Q.12 Answer:**

0

**Solution:**

**Expressing the determinant into product of two determinants**

$\Delta = \begin{vmatrix} x & y & 0 \\ \alpha & b & 0 \\ p & q & 0 \end{vmatrix} \begin{vmatrix} x & a & p \\ y & b & q \\ 0 & 0 & 0 \end{vmatrix} = 0$

**Q.13 Answer:**

None of these

**Solution:**

$\Delta = \begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix}$

$= \begin{vmatrix} 1+1+1 & \alpha+\beta+\gamma & \alpha^2+\beta^2+\gamma^2 \\ \alpha+\beta+\gamma & \alpha^2+\beta^2+\gamma^2 & \alpha^3+\beta^3+\gamma^3 \\ \alpha^2+\beta^2+\gamma^2 & \alpha^3+\beta^3+\gamma^3 & \alpha^4+\beta^4+\gamma^4 \end{vmatrix}$

**The above determinant can be expressed as product of two determinants. Thus,** $\Delta =$

$\begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} = [(\beta-\alpha)(\gamma-\alpha)(\gamma-\beta)]^2$

**Q.14 Answer:**

If $\mathrm{A}^3 + 2\ \mathrm{A}^2 + 3\ \mathrm{A} + 5\mathrm{I} = 0$; then $\mathrm{A}$ is invertible.

**Solution:**

**We have a theorem that if a square matrix A satisfies the equation** $a_0 + a_1x + a_2x^2 + \ldots\ldots + a_nx^n = 0$ **where** $a_0 \neq 0$ **then A is invertible. Since** $A, B$ **and** $C$ **are** $n \times n$ **matrices and A satisfies the equation** $x^3 + 2x^2 + 3x + 5 = 0$ **as** $\mathrm{A}^3 + 2\ \mathrm{A}^2 + 3\ \mathrm{A} + 5\mathrm{I} = 0$**, therefore, A invertible.**

**Q.15 Answer:**

$\mathrm{F}(-\alpha)$

**Solution:**

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$$F(\alpha)\cdot F(-\alpha)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos(-\alpha) & -\sin(-\alpha) & 0\\ \sin(-\alpha) & \cos(-\alpha) & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$F(\alpha)\cdot F(-\alpha)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos\alpha & \sin\alpha & 0\\ -\sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}\cos^2\alpha+\sin^2\alpha+0 & \cos\alpha\sin\alpha-\cos\alpha\sin\alpha+0 & 0+0+0\\ \sin\alpha\cos\alpha-\sin\alpha\cos\alpha+0 & \sin^2\alpha+\cos^2\alpha+0 & 0+0+0\\ 0+0+0 & 0+0+0 & 0+0+1\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=I \quad \left[\because \cos^2\alpha+\sin^2\alpha=1\right]$$

$$F(\alpha)\cdot F(-\alpha)=1 \qquad \therefore [F(\alpha)]^{-1}=F(-\alpha)$$

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